



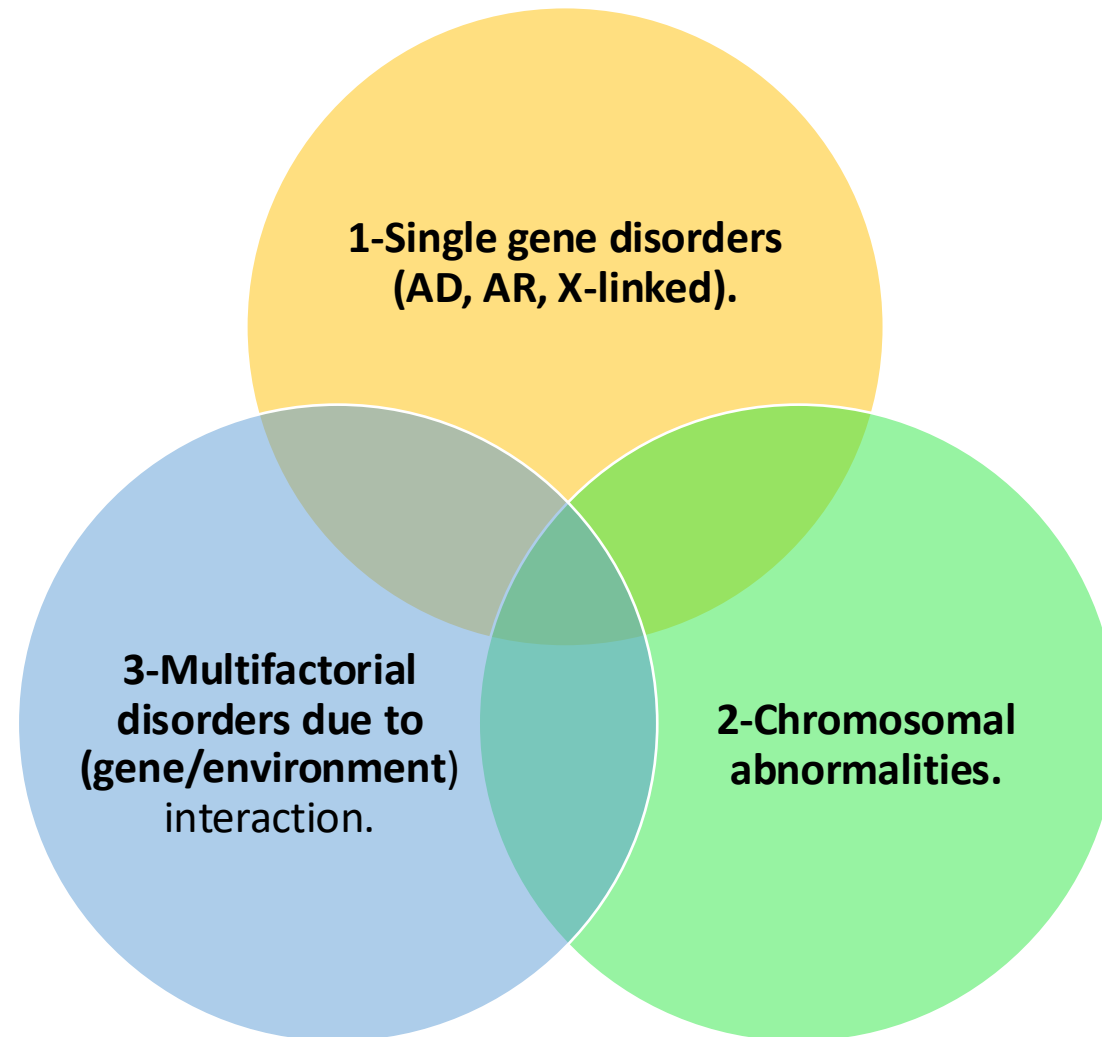
Dysmorphology

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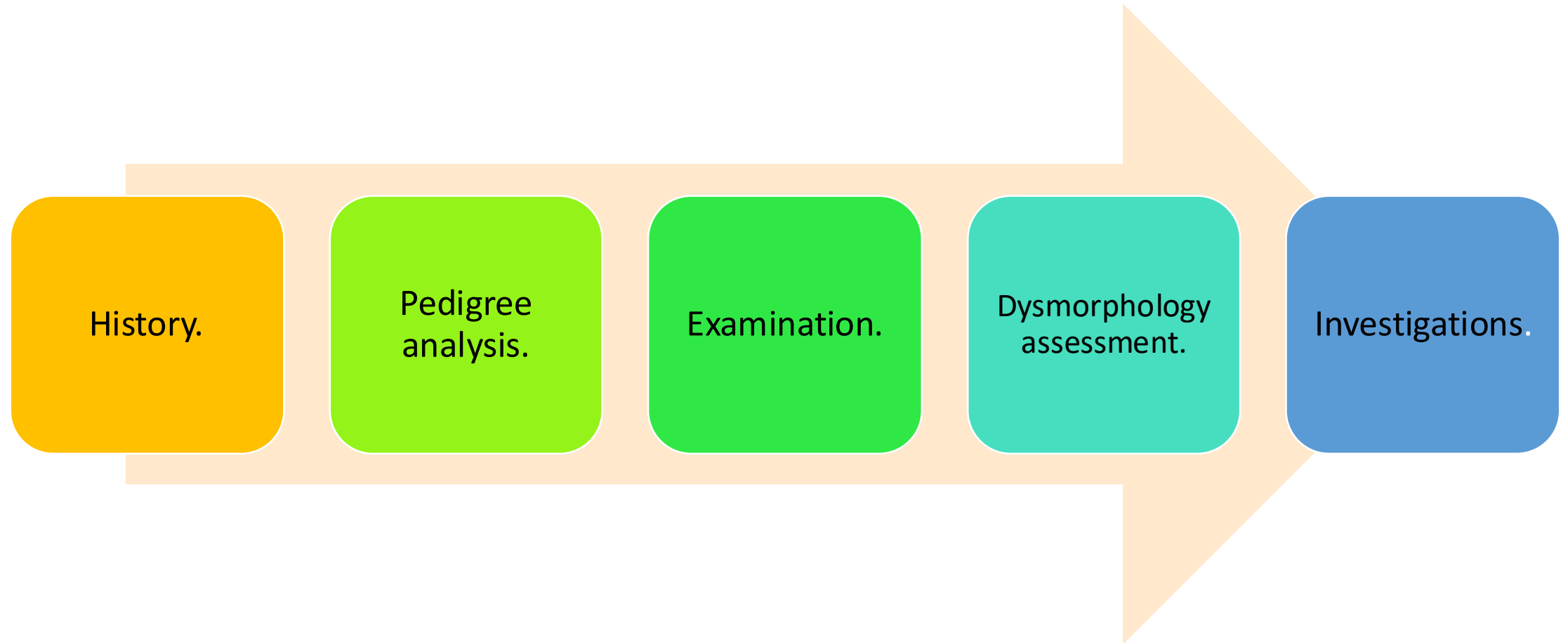
Lecture Learning Objectives

- By the end of this lecture the student will be able to ...
 1. Understand the **classification of genetic disorders** and their significance in clinical practice.
 2. Define **Dysmorphology** and understand its significance in clinical practice.
 3. Recognize the **definitions and characteristics of common dysmorphic features**.
 4. **Gain basic in taking personal and family medical histories** related to genetic conditions.
 5. Learn to perform pedigree analysis for assessing hereditary patterns.
 6. **Acquire techniques for thorough clinical examination** in the context of genetic disorders.
 7. Enhance the **ability to conduct dysmorphology assessments** for identifying physical anomalies.
 8. Familiarize with various **investigative methods** used in the diagnosis of genetic disorders.

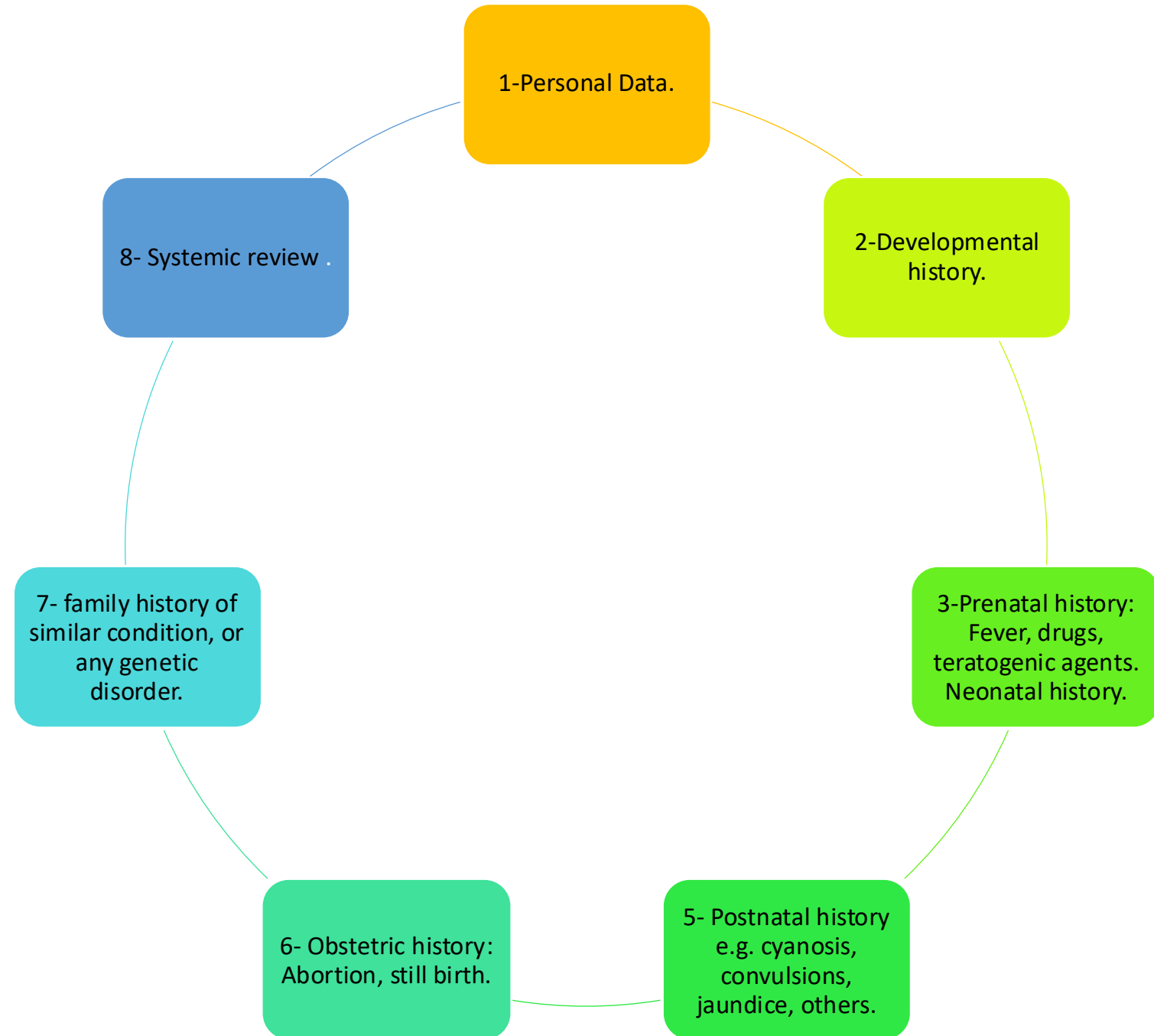
Classification of the genetic disorders



Clinical Approach to a Genetic Disorder



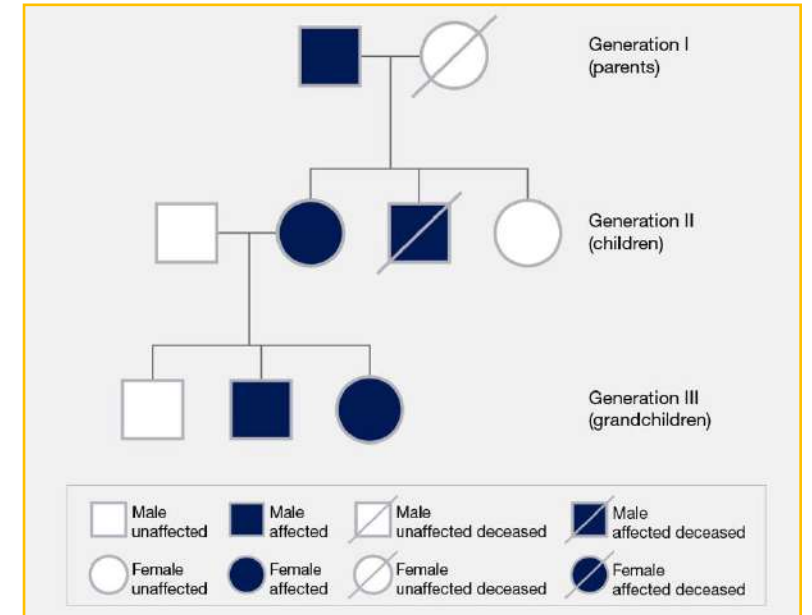
History



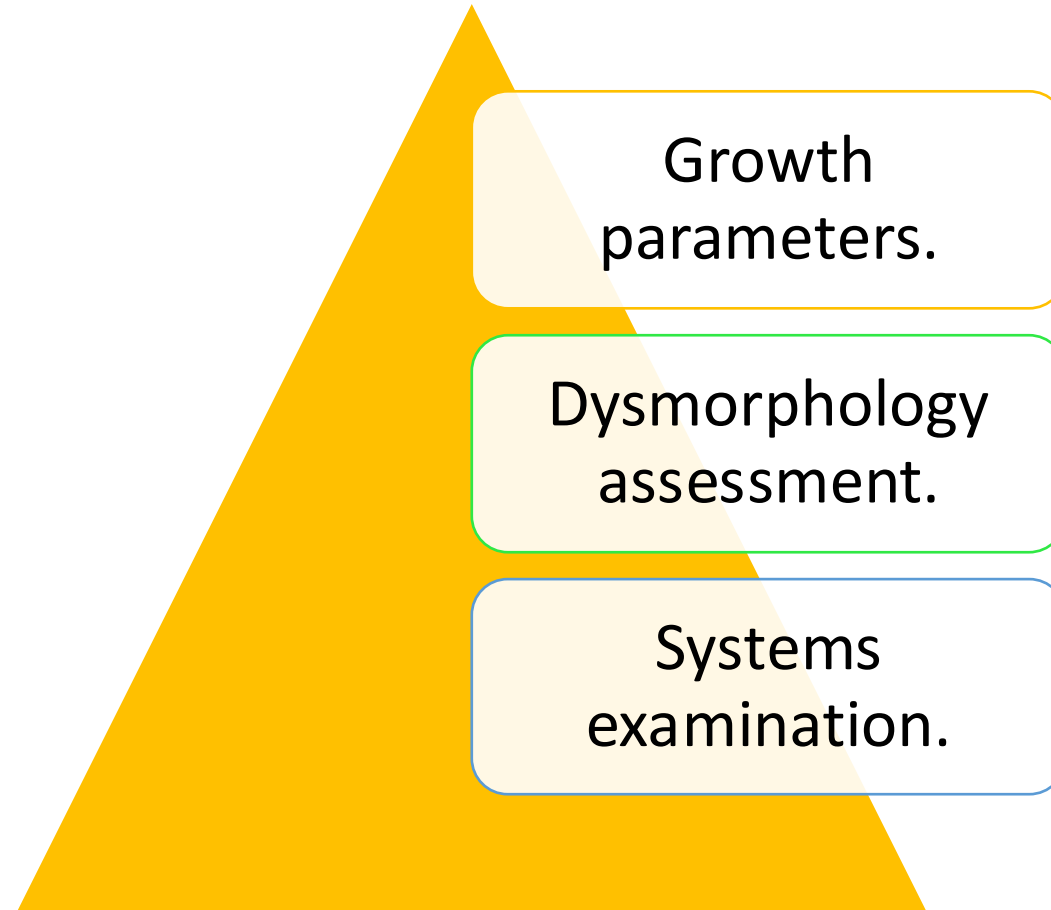
Pedigree Analysis

One of the most commonly used tools in medical genetics

- Illustrates the relationships among family members.
- Shows which family members are affected or unaffected by a genetic disease.
- Define the degrees of relationship.

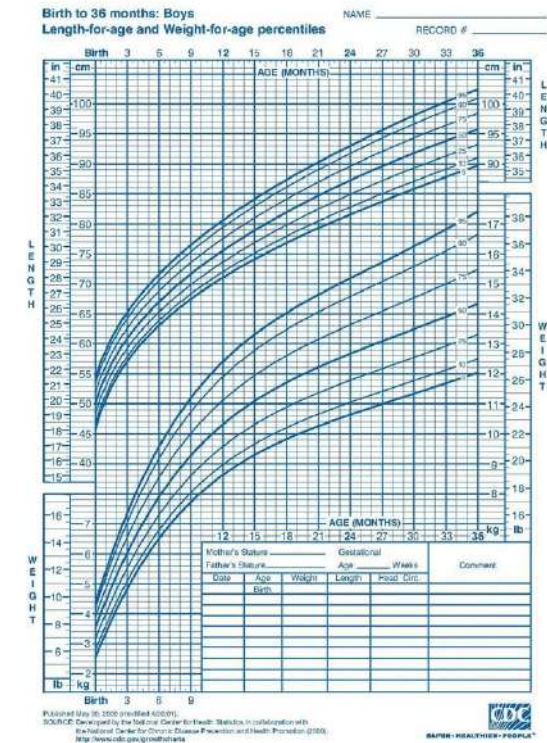
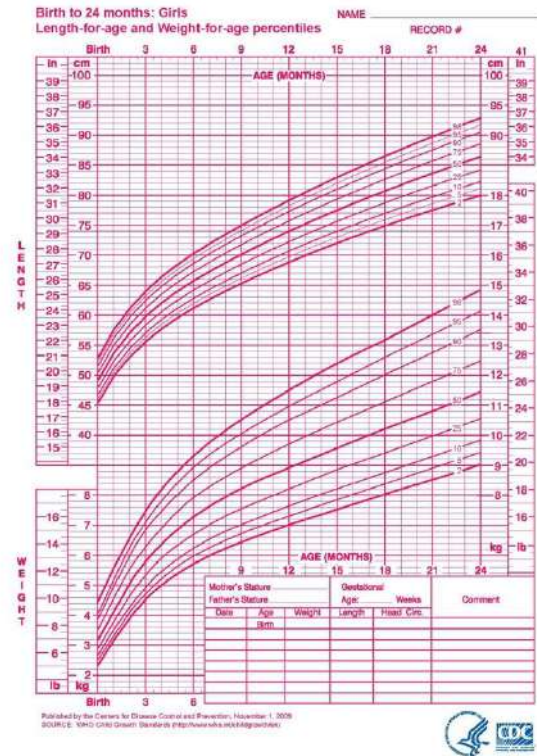


Examination:



Growth parameter

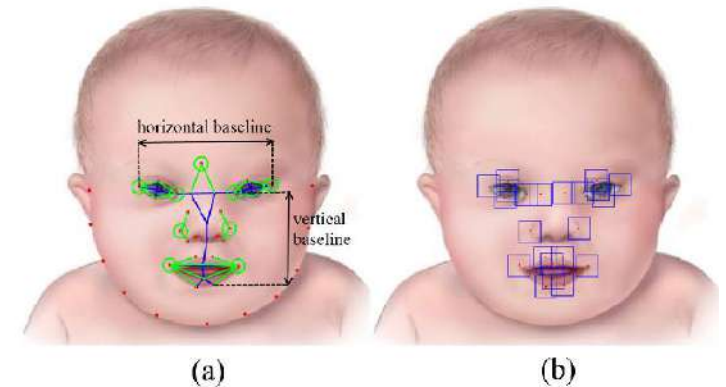
- Height
- Weight
- Head Circumference



Dysmorphology assesmnet

Dysmorphology :

- It is the study of human congenital malformations(birth defects) particularly those affecting the morphology (the anatomy) of the individual
- **Dys** (Greek)=disordered or abnormal and **Morph**=shape



Head size

Microcephaly

- **Definition:** Occipito-frontal (head) circumference (OFC) less than 3rd centile compared to appropriate, age matched, normal standards
- (HC : below -3 SD)
Microcephaly is a head circumference
- more than 3 standard deviations below the mean for age and gender.

HEAD



Macrocephaly

- **Definition:** Macrocephaly is a condition in which the human head is abnormally large. It may be pathological or benign, even a familial genetic characteristic.
- Definition: Occipitofrontal (head) circumference greater than 97th centile compared to appropriate, age matched, sex-matched normal standards

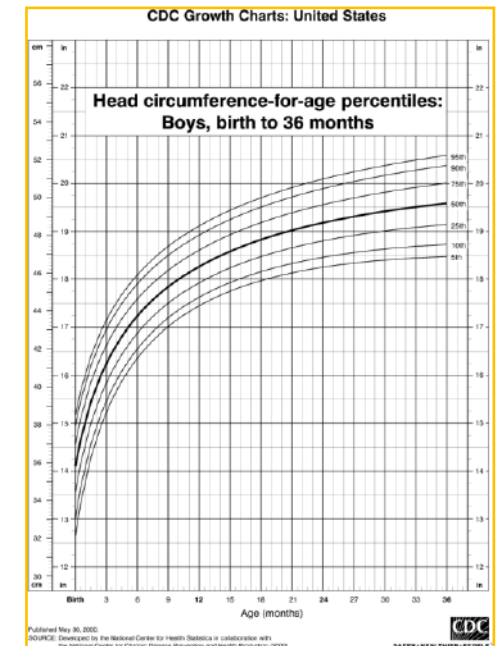




FIG. 7. *Macrocephaly*. Note the increased size of the cranium. Differences in size are difficult to appreciate but increased head size in this child is notable because of comparison with the smaller face.

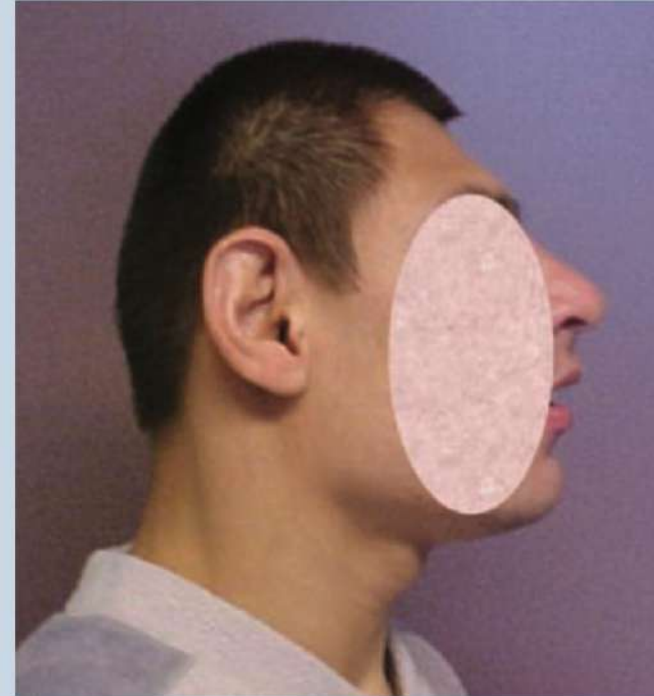
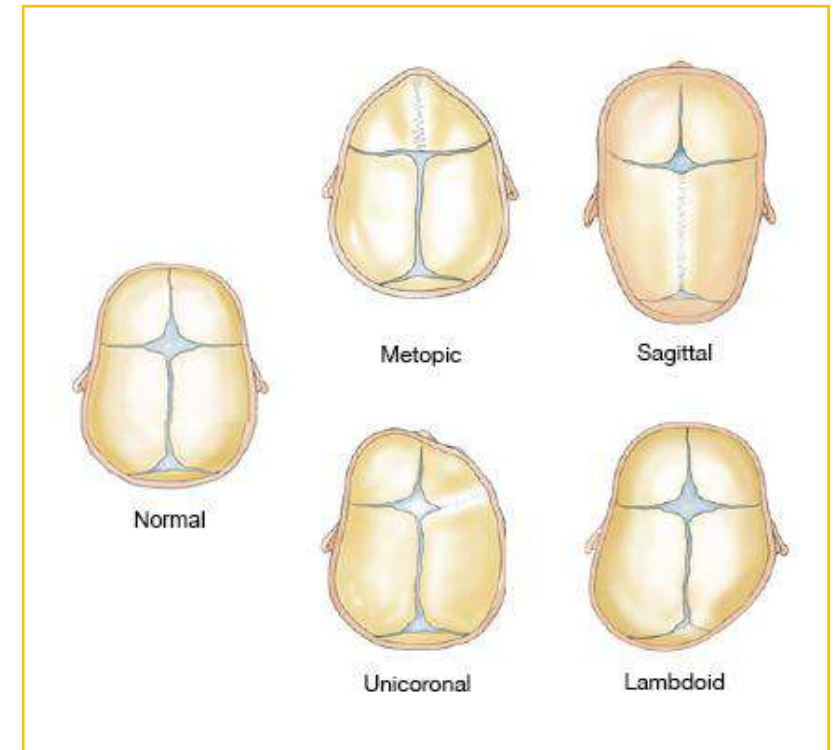
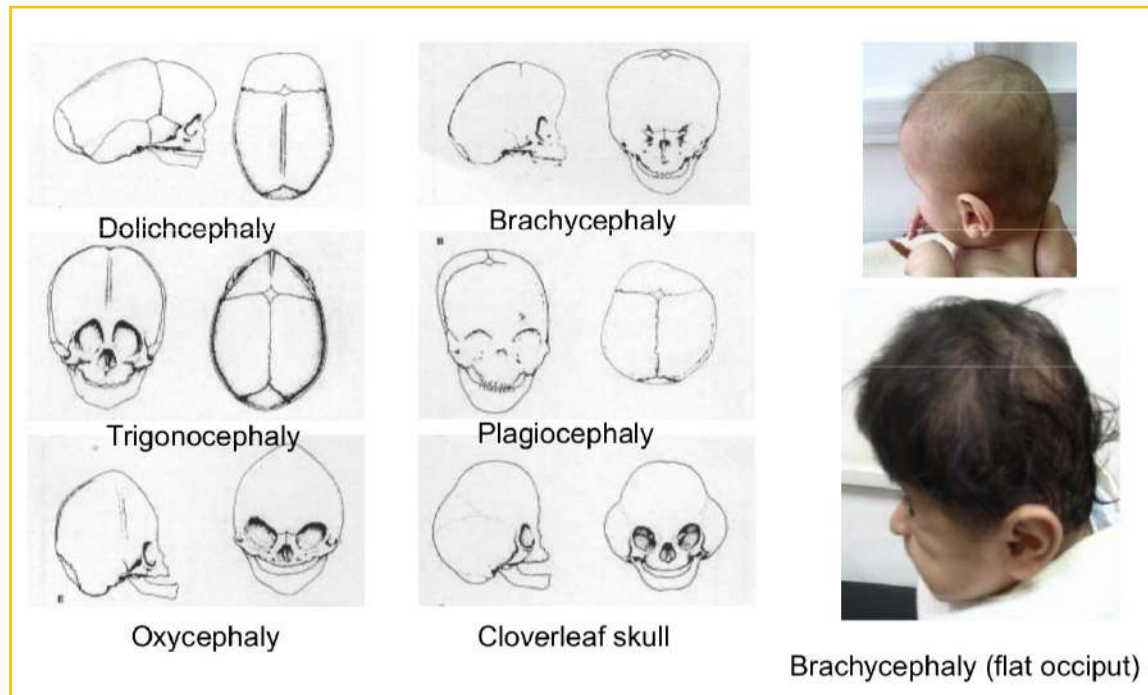


FIG. 8. *Microcephaly*. Decreased size of the cranium is accompanied by marked posterior sloping of the forehead.

Head shape:



Brachycephaly

- Definition: Cephalic index greater than 81% (Fig. 5). objective OR Apparently shortened anteroposterior dimension (length) of the head compared to width.



FIG. 5. *Brachycephaly*: The skull has a reduced anteroposterior dimension with the back of the head appearing to have reduced convexity.

Dolichocephaly

- Definition: Cephalic index less than 76% objective OR Apparently increased antero-posterior length of the head compared to width.



FIG. 6. *Dolichocephaly*. The skull has an increased antero-posterior dimension. Scaphocephaly is demonstrated on the right. Note that this subtype of dolichocephaly is “boat-shaped” with pointed anterior and posterior aspects of the cranial vault.

Plagiocephaly

Definition: Asymmetric head shape, which is usually a combination of unilateral occipital flattening with ipsilateral frontal prominence, leading to rhomboid cranial shape.



Trigonocephaly

Definition: Wedge-shaped, or triangular head, with the apex of the triangle at the midline of the forehead and the base of the triangle at the occiput .

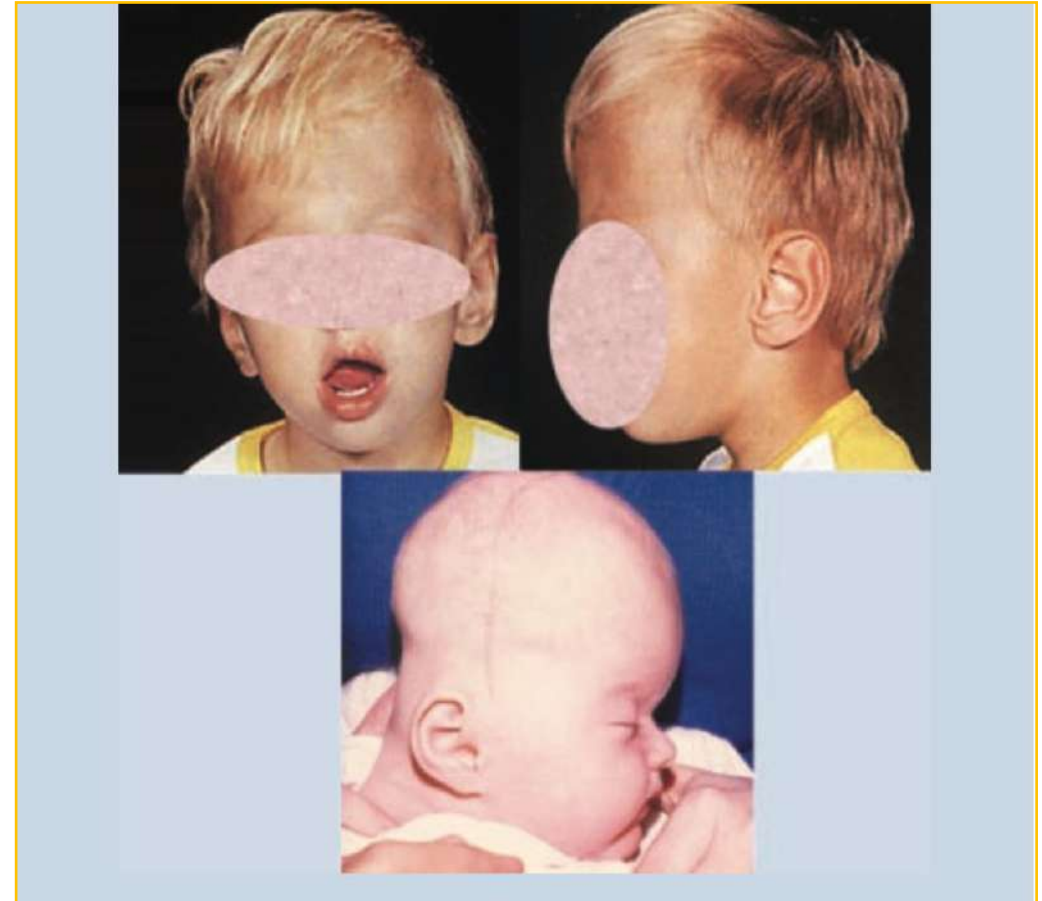
Comments: This shape should be assessed from above, with the examiner looking down on the head of the patient.



FIG. 13. *Trigonocephaly*. Note the wedge-shaped, or triangular head, with the apex of the triangle at the midline

Turricephaly

Definition: Tall head relative to width and length .



SCALP HAIR

Frontal Balding

Definition: Absence of hair in the anterior midline and/or parietal areas.

Hair, Frontal Upsweep

Definition: Upward and/or sideward growth of anterior hair



FIG. 15. *Frontal balding*. Note the absence of hair in the anterior midline and/or parietal areas.



FIG. 16. *Hair, frontal upsweep*. Note the pattern of upward and sideward growth of anterior hair.

SCALP HAIR

Hair Whorl, Abnormal Number

Definition: More than two clockwise hair whorls.



FIG. 18. *Hair whorl, abnormal position.* The hair whorl is positioned postero-inferiorly than its usual location lateral to the midline and close to the vertex of the skull.



FIG. 17. *Hair whorl, double.*

Hairline

Hairline, High Anterior

Definition: Distance between the hairline (trichion) and the glabella (the most prominent point on the frontal bone above the root of the nose), in the midline, more than two SD above the mean .

Apparently increased distance between the hairline and the glabella. subjective



FIG. 19. *Hairline, high anterior.* The high anterior hairline contributes to an appearance of tall forehead.

Hairline,

Hairline, Low Anterior

Definition: Distance between the hairline (trichion) and the glabella (the most prominent point on the frontal bone above the root of the nose), in the midline, more than two SD below the mean (Fig. 20a,b).



FIG. 20. *Hairline, low anterior.* The low anterior hairline contributes to an appearance of short forehead.

Hairline

Hairline, Low Posterior

Definition: Hair on the neck extends more inferiorly than usual (Fig. 21).



FIG. 21. *Hairline, low posterior.* Hair on the neck extends more inferiorly than usual, particularly in the lateral aspects.

Scalp Hair

Scalp Hair, Sparse

Definition: Decreased number of hairs per unit area

Widow's Peak

Definition: Frontal hairline with bilateral arcs to a low point in the midline of the forehead



FIG. 22. *Scalp hair, sparse.* Hair density is reduced giving a thinned appearance.



FIG. 23. *Widow's peak.* Frontal hairline shows bilateral arcs to a low point in the midline of the forehead.

Forehead

Forehead, Broad

Definition: Width of the forehead or distance between the fronto- temporales is more than 2 SD above the mean (Fig. 33). objective OR Apparently increased distance between the two sides of the forehead.

Forehead, Narrow

Definition: Width of the forehead or distance between the fronto- temporales is more than two SD below the mean (Fig. 34). objective OR Apparently narrow inter-temporal region.



FIG. 33. *Forehead, broad.* Note the increased distance between the two sides of the forehead.



FIG. 34. *Forehead, narrow.* Note the decreased distance between the two sides of the forehead with narrowing at the temples.

Forehead, Prominent

Definition: Forward prominence of the entire forehead, due to protrusion of the frontal bone.



FIG. 35. *Forehead, prominent.* The entire forehead is prominent due to protrusion of the frontal bone.

Metopic Depression

Definition: Linear vertical groove in the midline of the forehead, extending from hairline to glabella (Fig. 41). subjective

Comments: There is no underlying bony defect (metopism).

Metopic Ridge, Prominent

Definition: Vertical bony ridge positioned in the midline of the forehead



FIG. 41. *Metopic depression*. There is a linear vertical groove in the midline of the forehead, extending from hairline to glabella.



FIG. 42. *Metopic ridge, prominent*. Note the vertical bony ridge in the midline of the forehead.

Midface

Midface Retrusion

Definition: Posterior positioning and/or vertical shortening of the infraorbital and perialar regions, or increased concavity of the face and/or reduced nasolabial angle .

Midface Prominence

Definition: Anterior positioning of the infraorbital and perialar regions, or increased convexity of the face, or increased nasolabial angle .



FIG. 51. *Midface prominence*. Note prominence of the infraorbital and perialar regions leading to more pronounced convexity of the face and increased nasolabial angle.



FIG. 52. *Midface retrusion*. Note underdevelopment of the infra-orbital and peri-alar regions leading to more pronounced concavity of the face and reduced nasolabial angle. This gives the appearance of prognathia.

Micrognathia

Definition: Apparently reduced length and width of the mandible when viewed from the front but not from the side

Prognathism

Definition: Anterior protrusion of the mandibular alveolar ridge beyond the vertical plane of the maxillary alveolar ridge, best appreciated in profile

Retrognathia

Definition: Posteriorly positioned lower jaw, which is set back from the plane of the face when viewed from the side but not from the front



FIG. 61. *Prognathism*. There is anterior protrusion of the mandible such that the alveolar ridge extends beyond the vertical plane of the maxillary alveolar ridge.



FIG. 60. *Micrognathia*. There is shortening and narrowing of the mandible and chin.



FIG. 62. *Retrognathia*. The lower jaw is set back from the plane of the face.

Neck, Long

Definition: Increased distance from the point where neck and shoulders meet to the inferior margin of the occipital bone.

Neck, Short

Definition: Decreased distance from the point where neck and shoulders meet to the inferior margin of the occipital bone.



FIG. 72. *Neck, long.* Note the increased distance from the point where neck and shoulders meet to the inferior margin of the occipital bone.



FIG. 73. *Neck, short.* Note the decreased distance from the point where neck and shoulders meet to the inferior margin of the occipital bone.

Neck Webbing

Definition: A paravertically oriented fold of skin on the postero- lateral aspect of the neck, usually extending from the mastoid region of the skull to the acromion, and best appreciated in frontal or posterior view

Nuchal Skin, Redundant

Definition: Excess skin around the neck, often lying in horizontal folds



FIG. 74. Neck webbing. Note the bilateral folds of skin on the posterolateral aspect of the neck.

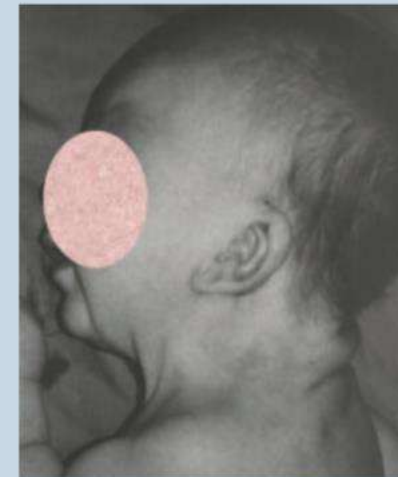


FIG. 75. Nuchal skin, redundant. Note the excess skin around the neck.

The Eye (Palpebral Fissure Slant)

- In the normal eye, the eyelids are generally positioned so that the lateral canthus is about 1 mm higher than the medial canthus.
- Slight upward or downward slanting of palpebral fissures normally occurs on a familial basis or in groups such as people of Asian descent.
- However, certain craniofacial syndromes frequently cause palpebral fissures to have a characteristic upward (Trisomy 21) or downward (Treacher Collins syndrome).



Upward slanting



Normal



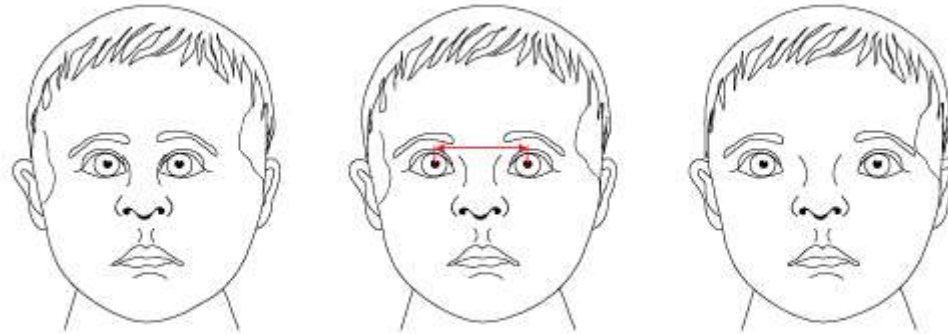
Downward slanting

Hypotelorism

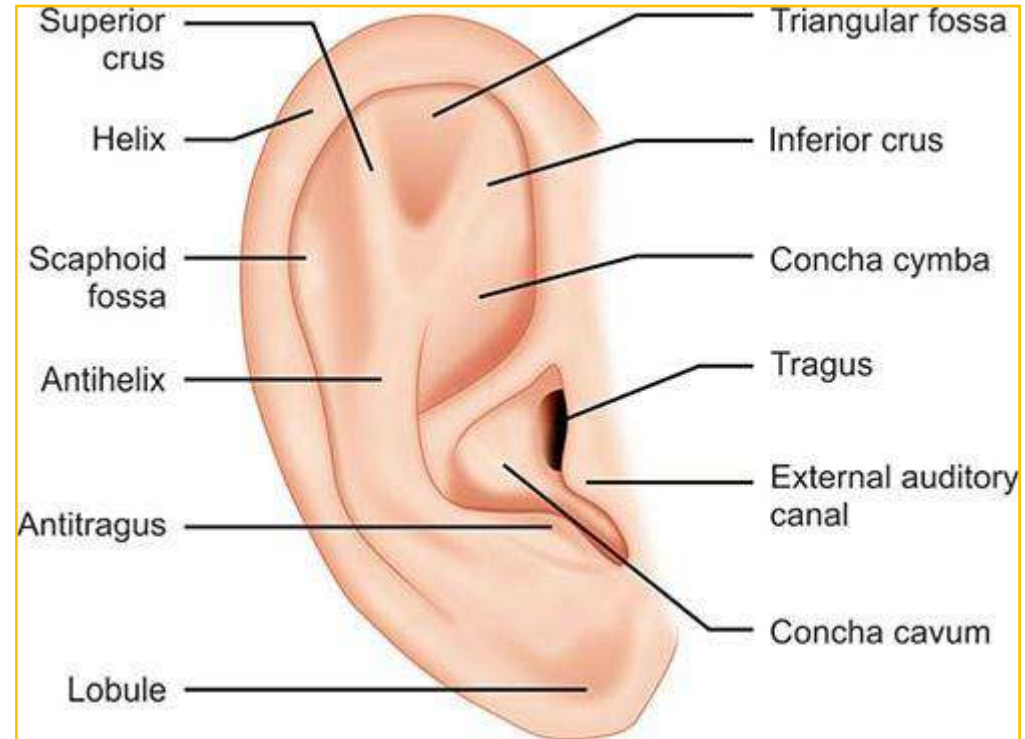
orbital hypotelorism meaning an abnormal decrease in the distance between the two eyes.

Hypertelorism

Orbital hypertelorism meaning an abnormal increase in the distance between the two eyes.



The Ear

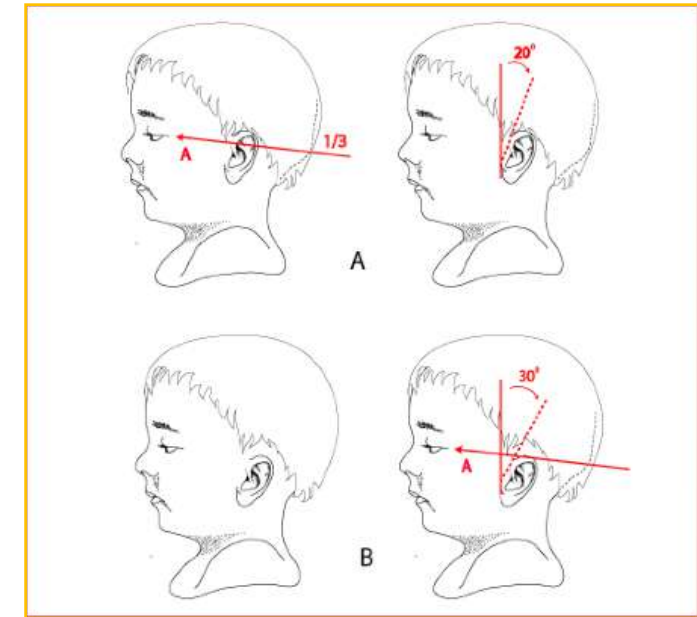


Ear position :

A: Normal Set Ear

B: Low Set & Posteriorly Rotated Ear

Low set ears: A minor anomaly involving an ear situated down below its normal location. The ear is low-set when the helix (of the ear) meets the cranium at a level below that of a horizontal plane through both inner canthi (the inside corners of the eyes).



Low-set, posteriorly rotated ear



Low-set ears

Anotia

- Definition: Complete absence of any auricular structures



FIG. 2. Anotia.

Antihelical Shelf

Definition: Antihelix protrusion directed more anteriorly than laterally, forming a shelf overlying the posterior concha



FIG. 4. Antihelical shelf.

Antihelix, Absent

Definition: No discernible ridge between concha and triangular fossa, and helix

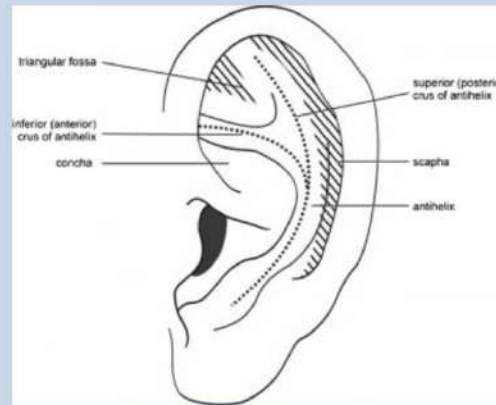


FIG. 3. Anatomy of antihelix.



FIG. 5. Absent antihelix.

Antihelix, Additional Crus

Definition: Supernumerary ridge or crus of the ear arising from the antihelix



FIG. 6. Additional crus of antihelix (Stahl ear). The additional crus is indicated in each panel by an arrow.

Antihelix, Inferior Crus, Broad

Definition: Increased width of the inferior cross-section of the inferior crus



FIG. 8. Variation in width of the inferior crus of the antihelix. a: Narrow. b: Average. c: Broad.

Antihelix, Stem, Prominent

Definition: Increased protrusion of the antihelical ridge, proximal to its bifurcation, relative to the prominence of the helix

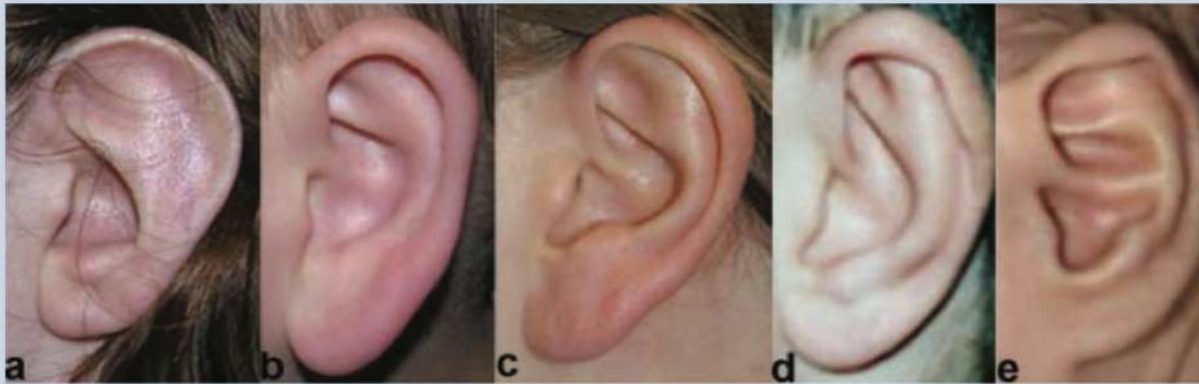


FIG. 10. Variation in protrusion of the antihelix stem. a: Marked underdevelopment. b: Mild underdevelopment. c: Average development. d: Mild prominence. e: Marked prominence.

Antihelix, Stem, Serpiginous

- Definition: Posterior curving of the antihelix from its origin at the antitragus, traveling initially almost perpendicular to the descending helix and obscuring some of the concha .



FIG. 11. Serpiginous stem.

Antitragus, Absent

- Definition: Absence of the anterosuperior prominence of the area between the bottom of the incisura and the inner margin of the antihelix



FIG. 13. Normal anatomy of antitragus.

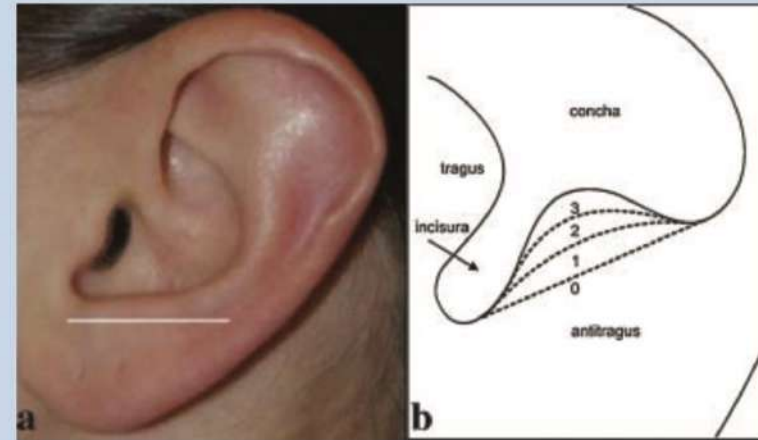


FIG. 14. Variation in size of the antitragus. a: Absent antitragus. b: The size of the antitragus is divided into 4°, of which 0 indicates absence, 1 is underdeveloped, 2 indicates the average size, and 3 indicates prominence (Fig. 14b adapted from Lange, 1966).

Ear, Crumpled

- Definition: Distortion of the course of the normal folds of the ear and the appearance of supernumerary crura and folds



FIG. 21. Crumpled ear.

Ear, Cupped

Definition: Laterally protruding ear that lacks antihelical folding (including absence of inferior and superior crura)



FIG. 22. Cupped ear.

Cryptotia

- Definition: Invagination of the superior part of the auricle under a fold of temporal skin

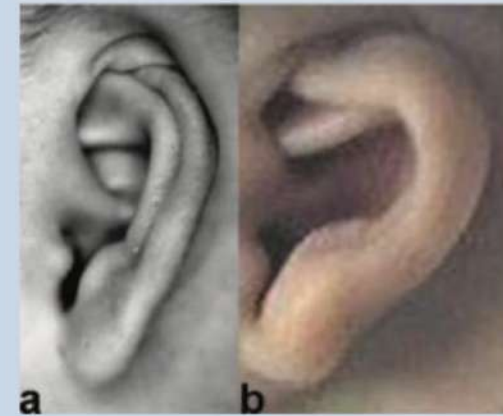


FIG. 20. Cryptotia. a. Type I; b. Type II.

Helix, overfolded

Definition: Excessive curling of the helix edge, whereby the free edge is parallel to the plane of the ear



FIG. 42. Overfolded helix.

Helix, Underfolded

Definition: Underdevelopment of the helix that either affects the entire helix, or is localized



FIG. 45. Localized underdeveloped helix.

Helix, Posterior Pit

Definition: Permanent indentation on the posteromedial aspect of the helix that may be sharply or indistinctly delineated



FIG. 43. Pits in posterior helix.

Lobe, Absent

Definition: Absence of fleshy non-cartilaginous tissue inferior to the tragus and incisura



FIG. 47. Absent lobe.

Lobe, Attached

Definition: Attachment of the lobe to the side of the face at the lowest point of the lobe without curving upward



FIG. 49. Attached lobe.

Microtia, First Degree

Definition: Presence of all the normal ear components and the median longitudinal length more than 2 SD below the mean

Microtia, Second Degree

Definition: Median longitudinal length of the ear more than 2 SD below the mean in the presence of some, but not all, parts of the normal ear

Microtia, Third Degree

Definition: Presence of some auricular structures, but none of these structures conform to recognized ear components



FIG. 56. Microtia, second degree.

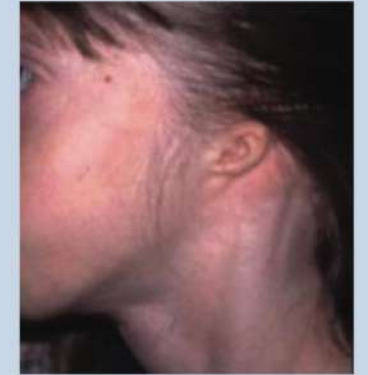


FIG. 55. Microtia, first degree.



FIG. 57. Microtia, third degree.

Tag, Auricular

Definition: Small protrusion within the pinna



FIG. 60. Pretragial ectopias; in (a) note the resemblance of the ectopia to normal helix; in (c) there is clear duplication of hyoid components in this example of polyotia. Note the hair on part of the duplication in (b), which is typical of the infant helix but not the tragus. Note also that neither figure shows presence of a normal tragus. ((c) Courtesy of Dr Sergio B. De Sousa).

- Microtia: Underdeveloped outer ear.
- Anotia: Missing one or both ears.
- Protruding ears: More than 2 cm from the head.



Microtia (small or dysplastic ears)



Mouth (Philtrum)

- the vertical groove between the base of the nose and the border of the upper lip.

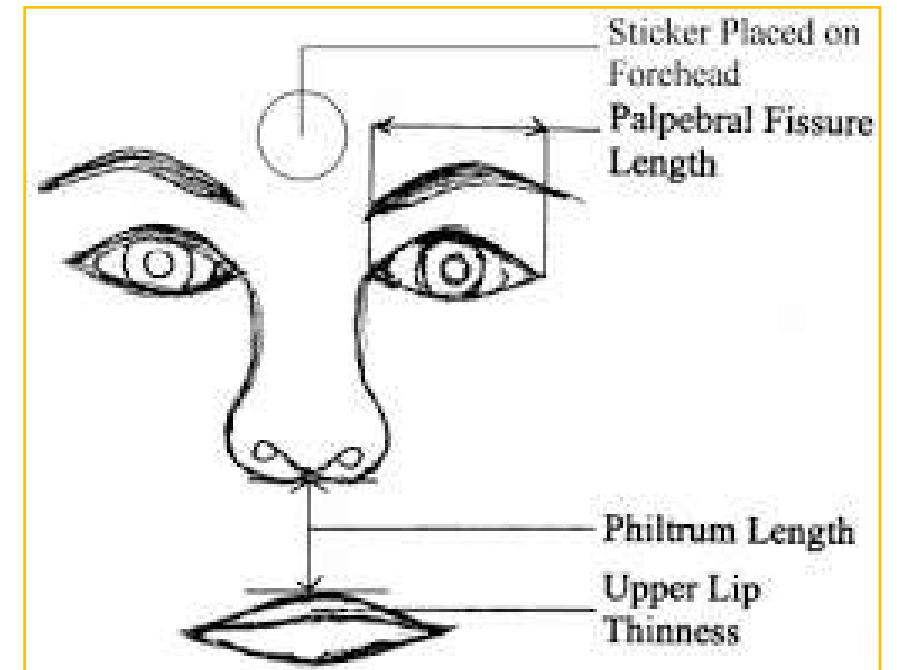
e.g. short, long or flat



Short philtrum



Long flat philtrum



Mouth (lips, tong, palate)

- Macrostomia (large mouth)
- Microstomia (small mouth)
- Macroglossia (large tongue)



Thin tented lips



Thick patulous lips



Cleft lip



Macrostomia



Lip pits



Microstomia



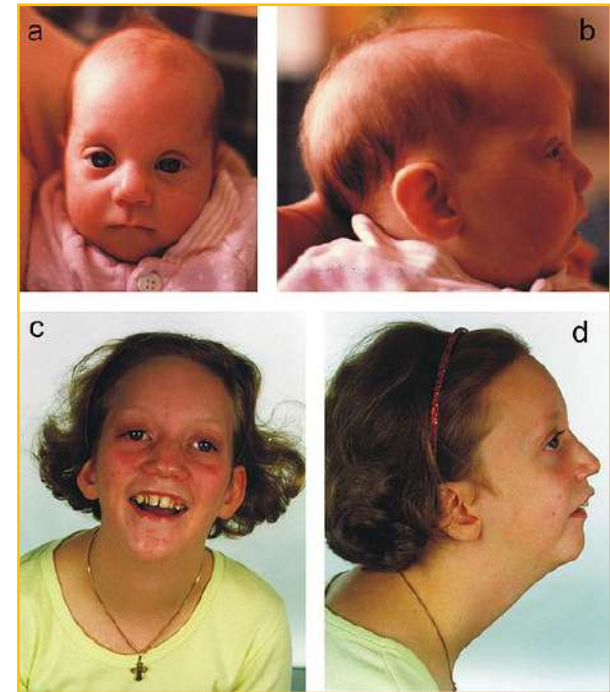
Macroglossia (large tongue) with bifid tip



Micrognathia

Mandible

- Small mandible: Micrognathia.
- Receding mandible: Retrognathia.
- Protrusion of mandible : Prognathism.



Hands

- **Arachnodactyly** ("spider fingers") is a condition in which the fingers and toes are abnormally long and slender, in comparison to the palm of the hand and arch of the foot.)
- **Brachydactyly** (short fingers & toes)



Fingers

- **Polydactyly** refers to extra fingers or toes (digits) that are present at birth.

Types of Polydactyly



Preaxial (Radial)

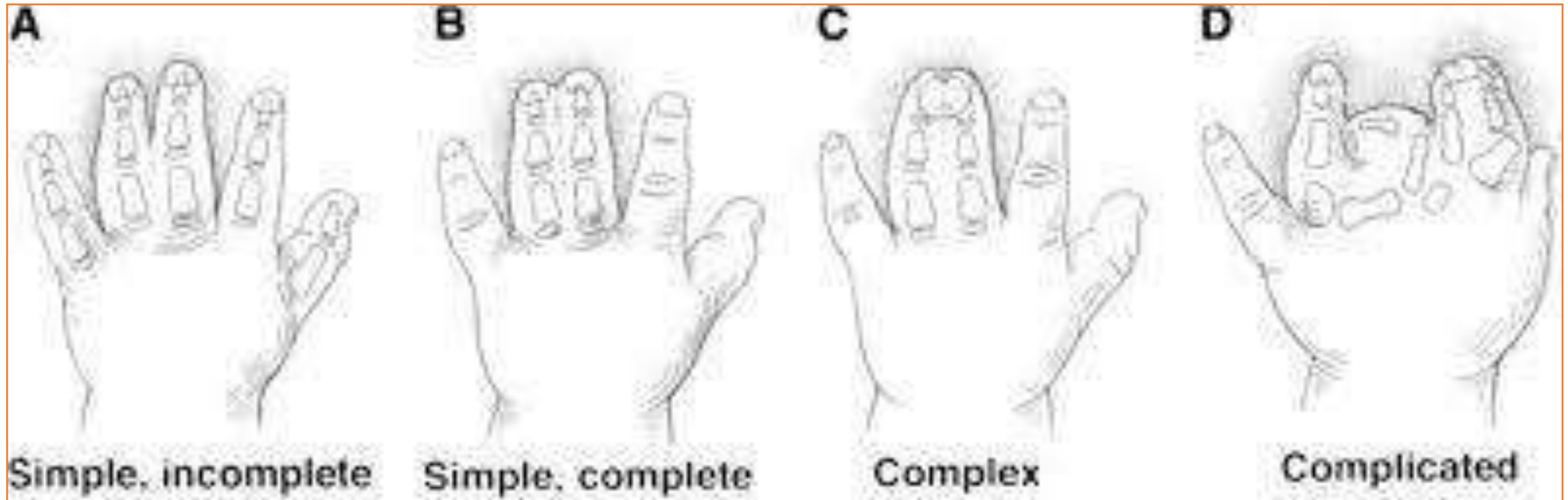


Central



Postaxial (Ulnar)

Syndactyly (Fused Digits) (Fingers or Toes)



Down Syndrome/Trisomy 21

Clinical Features:

Newborn period:

Hypotonia, excess nuchal skin

Craniofacial:

- Brachycephaly, microcephaly
- Epicanthic folds, upslanting palpebral fissures, depressed nasal bridge
- Protruding tongue
- Small/low-set ears

Limbs:

- Single palmar crease
- Clinodactyly of 5th finger
- Wide gap between 1st and 2nd toes

Congenital heart defects:

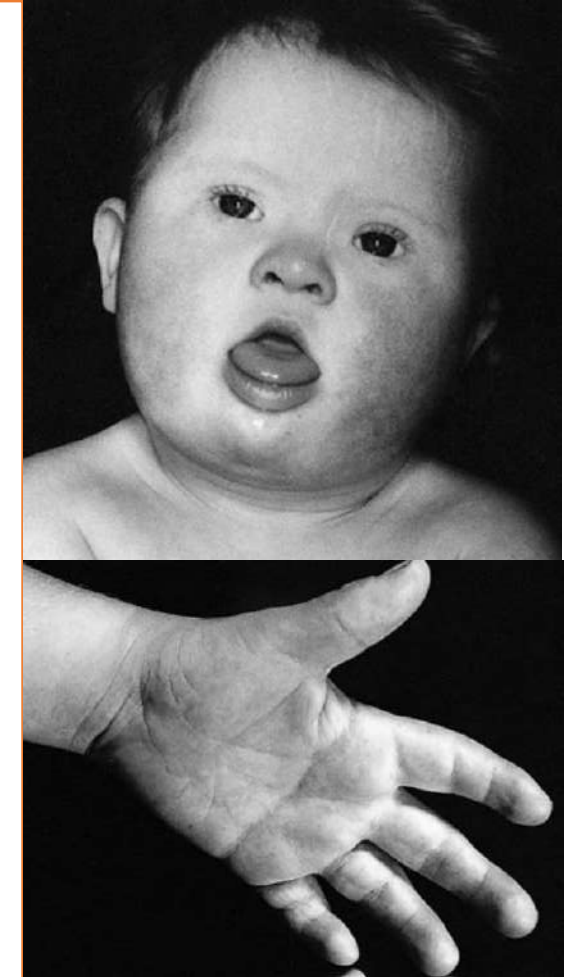
ASD, VSD, AV canal

GI:

Duodenal atresia, celiac disease, umbilical hernia

Other:

- Short stature, growth delay, intellectual disability
- Congenital hypothyroidism
- Strabismus, myopia, conductive hearing loss
- Leukemia (ALL)
- Alzheimer disease



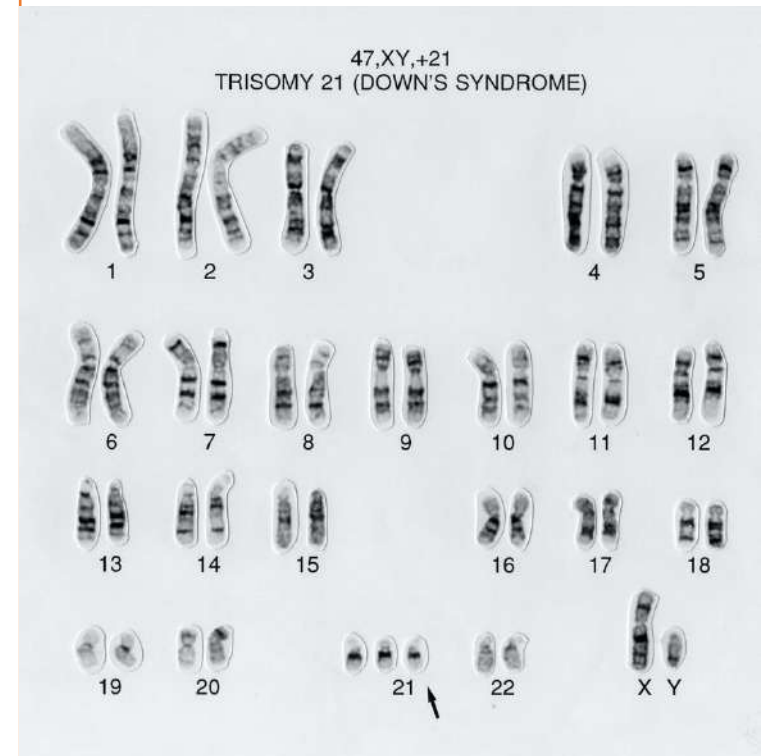
Down Syndrome/Trisomy 21

Inheritance:

- 95% de novo (non-disjunction mostly in maternal meiosis I)
- 4% due to Robertsonian translocation (1/3 with a carrier parent)
- 1% due to mosaicism

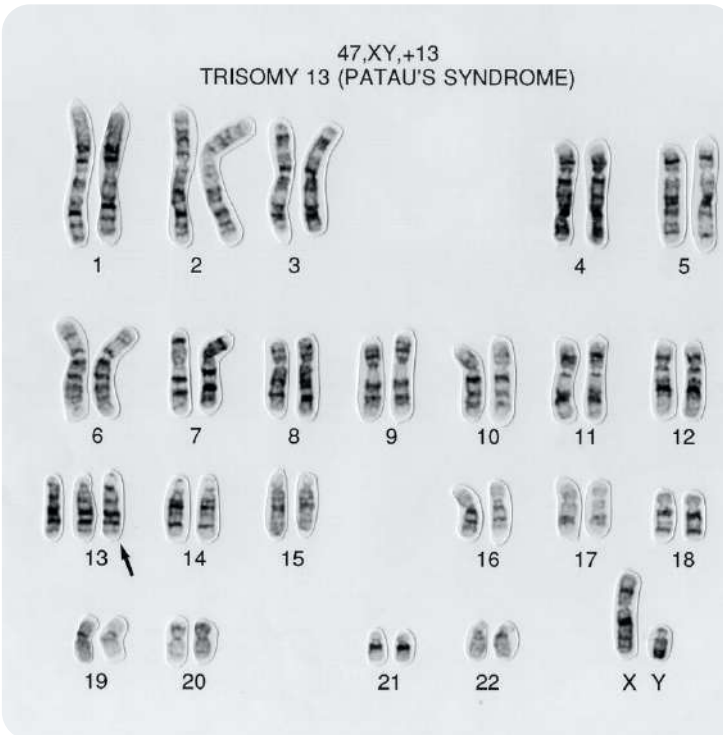
Recurrence risk (RR):

- Trisomy 21 due to non-disjunction → RR is **1%**, plus maternal age-related risk
- In familial translocation cases → RR vary from **1-3%** for **male** carriers and up to **10-15%** for **female** carriers
- Carriers of a 21q21q translocation → RR is **100%**



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Patau syndrome/Trisomy 13



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Clinical features:

- Holoprosencephaly, microcephaly, scalp defects
- Microphthalmia, low birth weight
- Polydactyly
- Midline cleft lip and palate
- Cardiac and renal anomalies
- Intellectual disability

Inheritance:

- 75% due to maternal nondisjunction
- 20% due to a translocation
- 5% due to mosaicism

Prognosis: very poor, with most infants dying during the first days or weeks of life



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Turner Syndrome

Responsible gene: *SHOX* (short stature homeobox), located Xp22.33

Inheritance: sporadic

Clinical features:

- Congenital lymphedema (swelling of hand and feet at birth)
- Short stature, growth failure
- Short webbed neck, low posterior hairline, increased carrying angles at the elbows, widely spaced nipples
- Normal intelligence
- Congenital heart defects: coarctation of the aorta, bicuspid aortic valve
- Ovarian failure/streak gonads leads to primary amenorrhea and infertility
- Other: hypothyroidism, renal malformation (horseshoe kidneys)

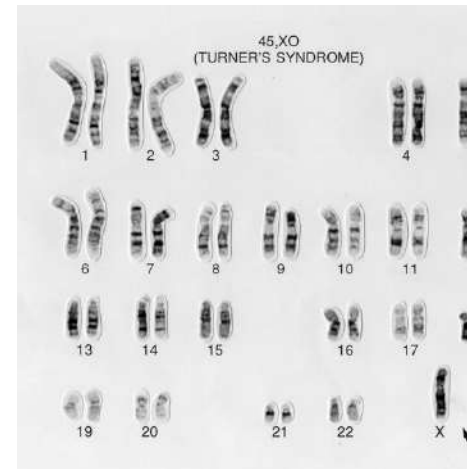
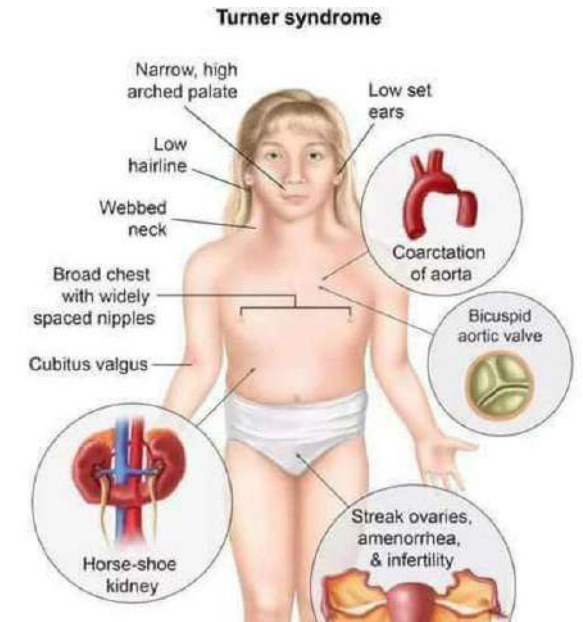


Table 17.6 Chromosome Findings in Turner Syndrome

Karyotype	Frequency (%)
Monosomy X: 45,X	50
Mosaicism (e.g., 45,X/46,XX)	20
Isochromosome: 46,X,i(Xq)	15
Ring: 46,X,r(X)	5
Deletion: 46,X,del(Xp)	5

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Emery's Elements of Medical Genetics 15th Edition

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Investigation:

- The investigations in a child with a genetic disease are defined by the anomalies encountered.
- 1-Chromosomal studies.
2-Metabolic studies.
3-Molecular studies.
4-The tests may also be more general, such as an X-ray, ultrasound, urine analysis, skin biopsy.

References

- •Emery and Rimoin's Principles and Practice of Medical Genetics and Genomics. 7th Edition 2019.
Edited by: Reed Pyeritz Bruce Korf Wayne Grody.
- •Emery's Elements of Medical Genetics 16th edition 2022 by Turnpenny and Ellard.
- Elements of dysmorphology.